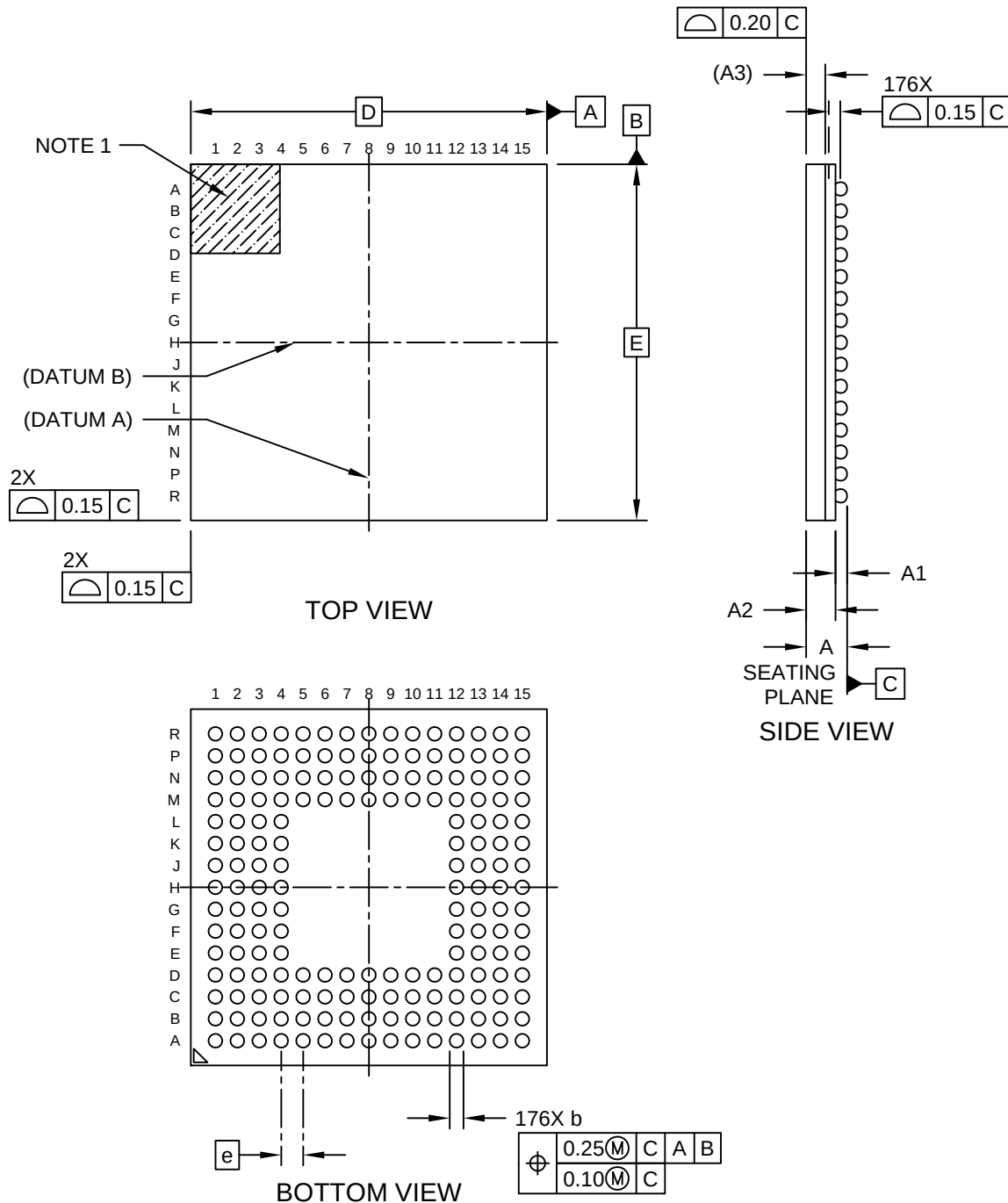


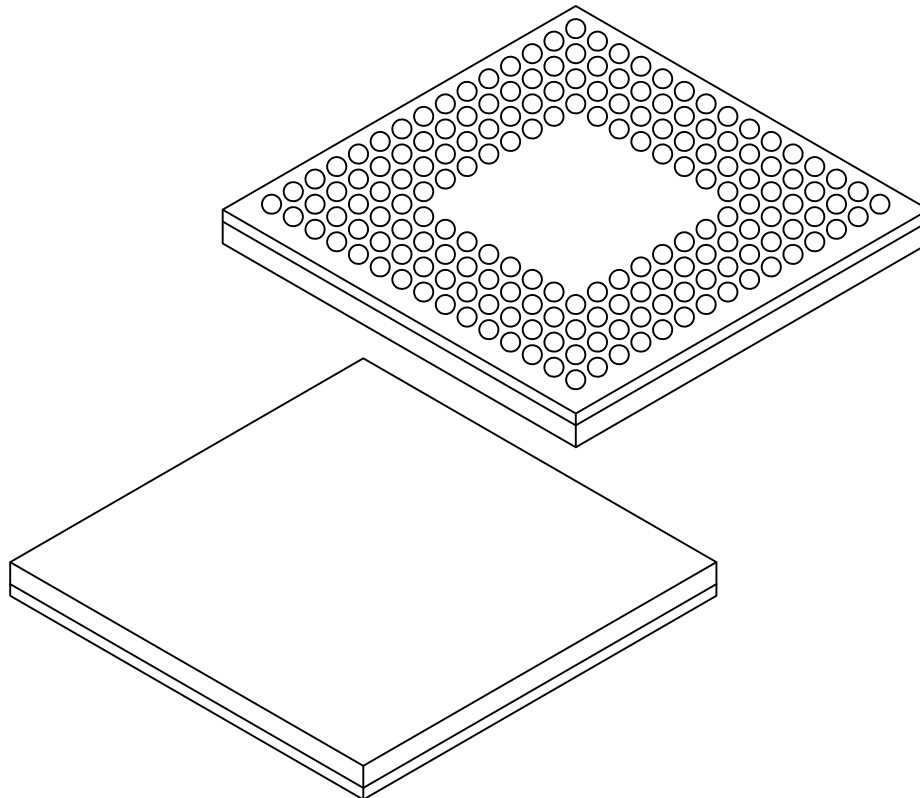
176-Ball Low Profile Fine Pitch Ball Grid Array (ALB) - 13x13x1.5 mm Body [LFBGA] Atmel Legacy Global Package Code CSX

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



**176-Ball Low Profile Fine Pitch Ball Grid Array (ALB) - 13x13x1.5 mm Body [LFBGA]
Atmel Legacy Global Package Code CSX**

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



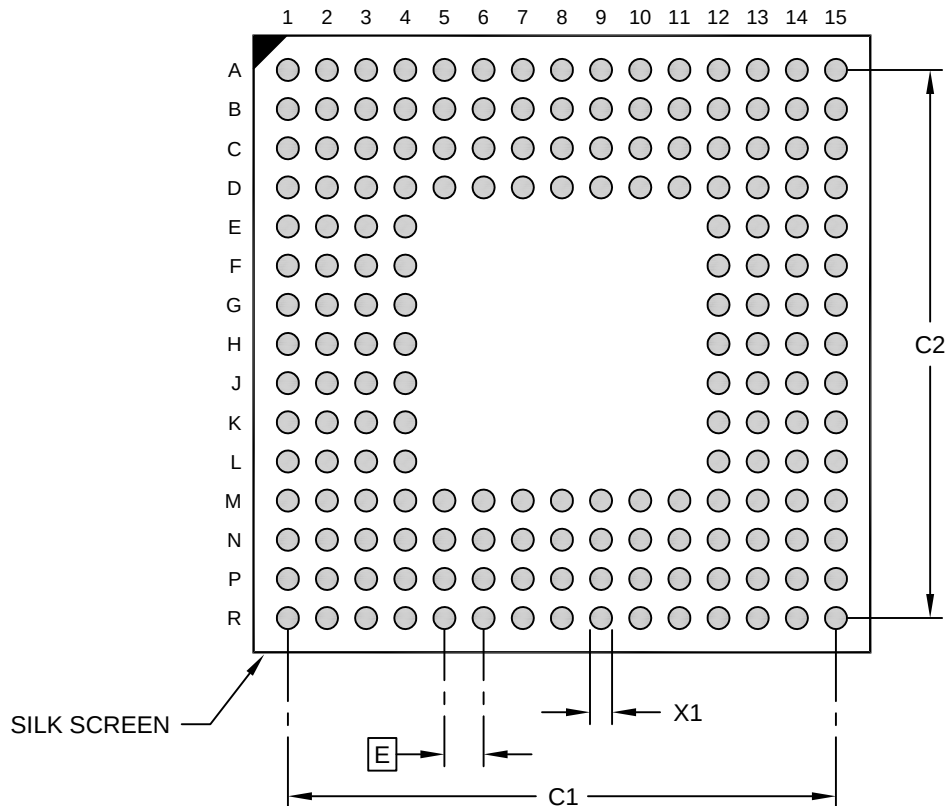
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	176		
Pitch	e	0.80 BSC		
Overall Height	A	-	-	1.50
Ball Height	A1	0.35	0.425	0.50
Molded Cap Height	A3	0.70 REF		
Overall Length	D	13.00 BSC		
Overall Width	E	13.00 BSC		
Terminal Diameter	b	0.45	0.50	0.55

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

176-Ball Low Profile Fine Pitch Ball Grid Array (ALB) - 13x13x1.5 mm Body [LFBGA] Atmel Legacy Global Package Code CSX

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.80 BSC		
Contact Pad Spacing	C1		11.20	
Contact Pad Spacing	C2		11.20	
Contact Pad Width (X20)	X1			0.45

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23123 Rev A